

Report Baby Names

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1. Introduction

A New York based toy design agency requested an analysis of historical baby naming trends in the United States between 1910 and 2014. The objective is to determine whether popular names persist over time, identify media-related naming spikes, and provide evidence-based recommendations for selecting toy character names likely to remain attractive to future consumers.

1.1. Data Sources

- US Baby Names by State (1910–2014)
- Billboard Top 100 Charts
- HBO Titles and Credits

1.2. Objectives

- Measure persistence of popular names over time.
- Identify major spikes in popularity.
- Investigate media influences on naming behaviour.
- Develop recommendations for toy character naming in the New York market.

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2. Persistence of Popular Baby Names

2.1. Spearman Rank Correlation Analysis

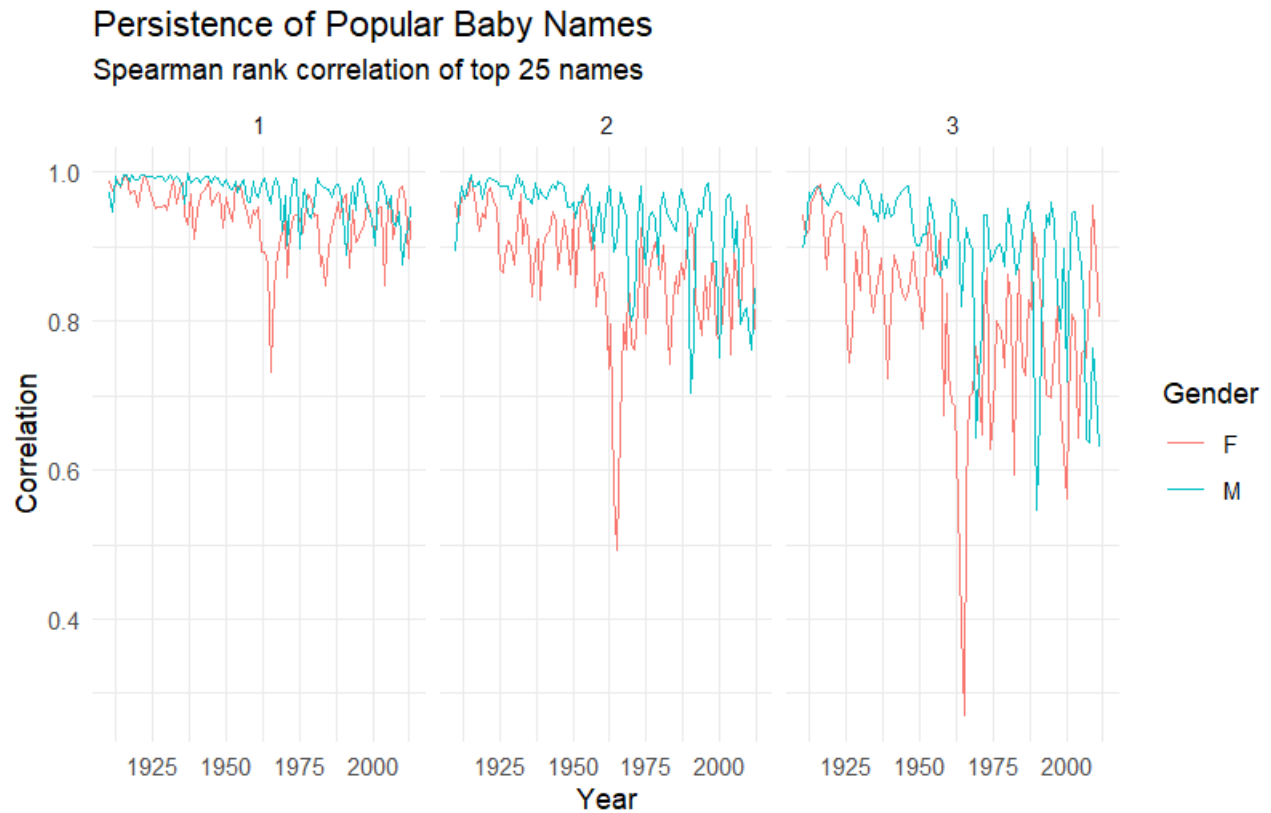


Figure 2.1: Persistence of popular baby names using Spearman rank correlations.

2.1.1. Key Findings

- Popular names were highly persistent during most of the twentieth century.
- Persistence declines as the comparison horizon increases from one to three years.
- Naming persistence decreases noticeably after the 1990s.
- Modern naming trends appear substantially more volatile.

2.1.2. Interpretation

Higher Spearman correlations indicate that the ranking of popular names remains stable through time. Lower correlations imply that popular names are replaced more rapidly by new trends.

3. Before vs After 1990

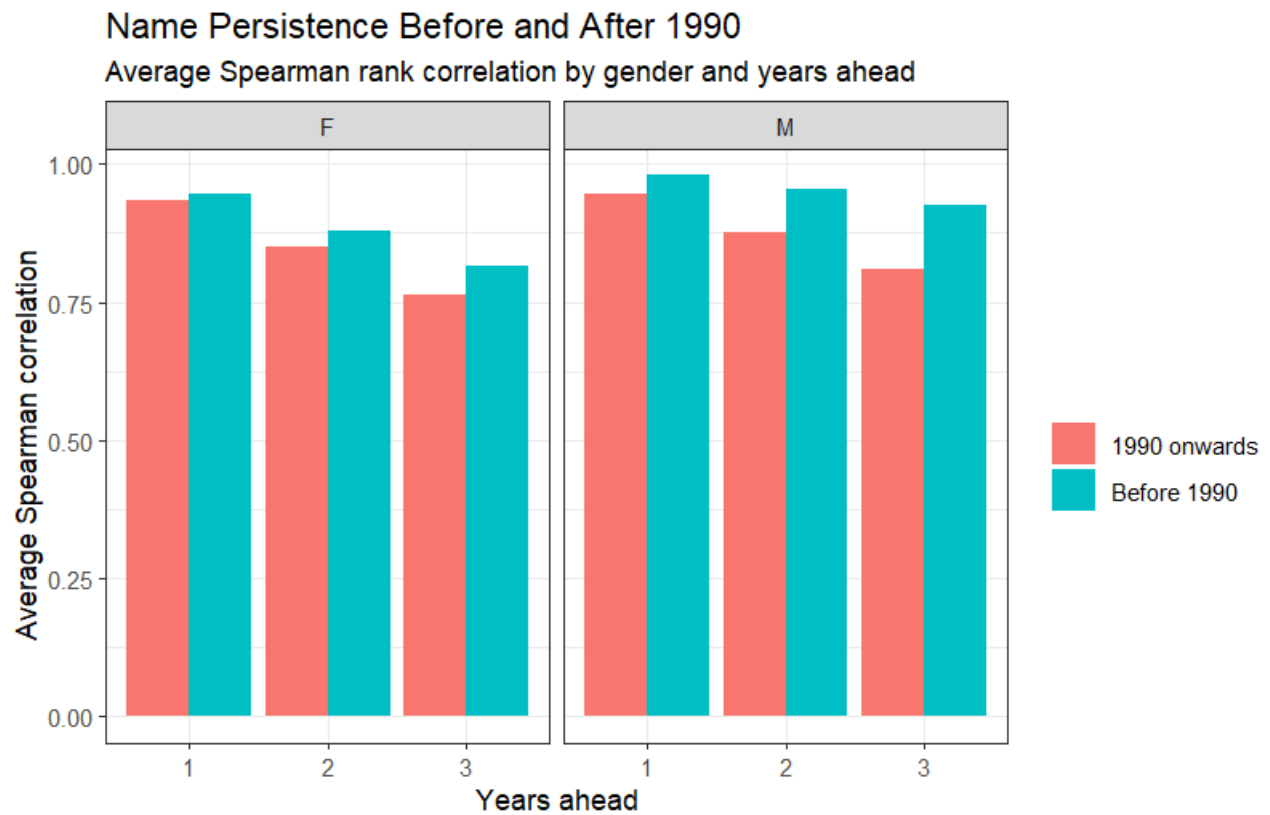


Figure 3.1: Average name persistence before and after 1990.

3.0.1. Key Findings

- Average Spearman correlations are consistently higher before 1990.
- Both boys' and girls' names exhibit reduced persistence after 1990.
- The agency's hypothesis is supported.

3.0.2. Business Implication

- Modern naming trends change more rapidly.
- Trend-driven toy names face a greater risk of becoming outdated.
- Stable names are preferable for long-term product lines.

4. Name Popularity Spikes

4.1. Largest Relative Name Spikes

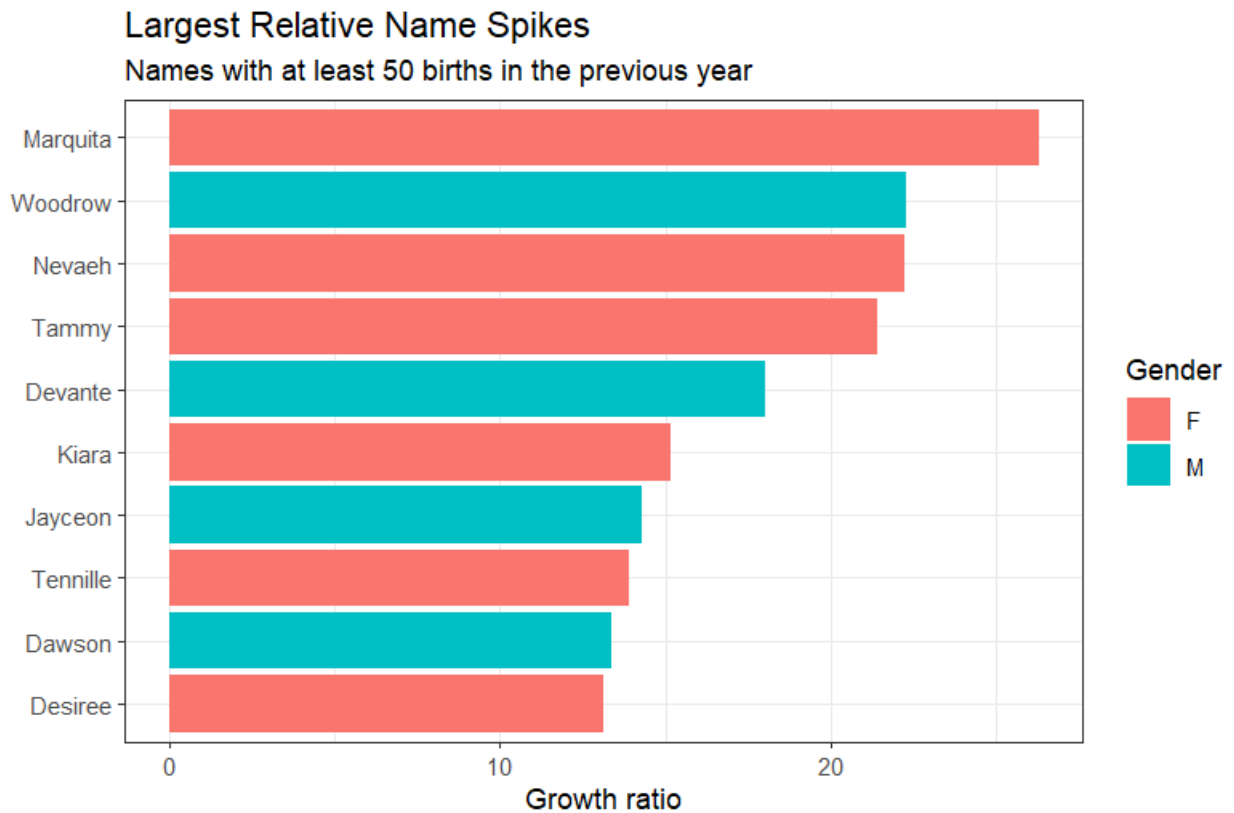


Figure 4.1: Largest relative increases in baby-name popularity.

4.1.1. Key Findings

- Several names experienced explosive year-on-year growth.
- Relative spikes typically occur among previously uncommon names.
- Many spikes appear linked to cultural and media influences.

4.2. Largest Absolute Name Spikes

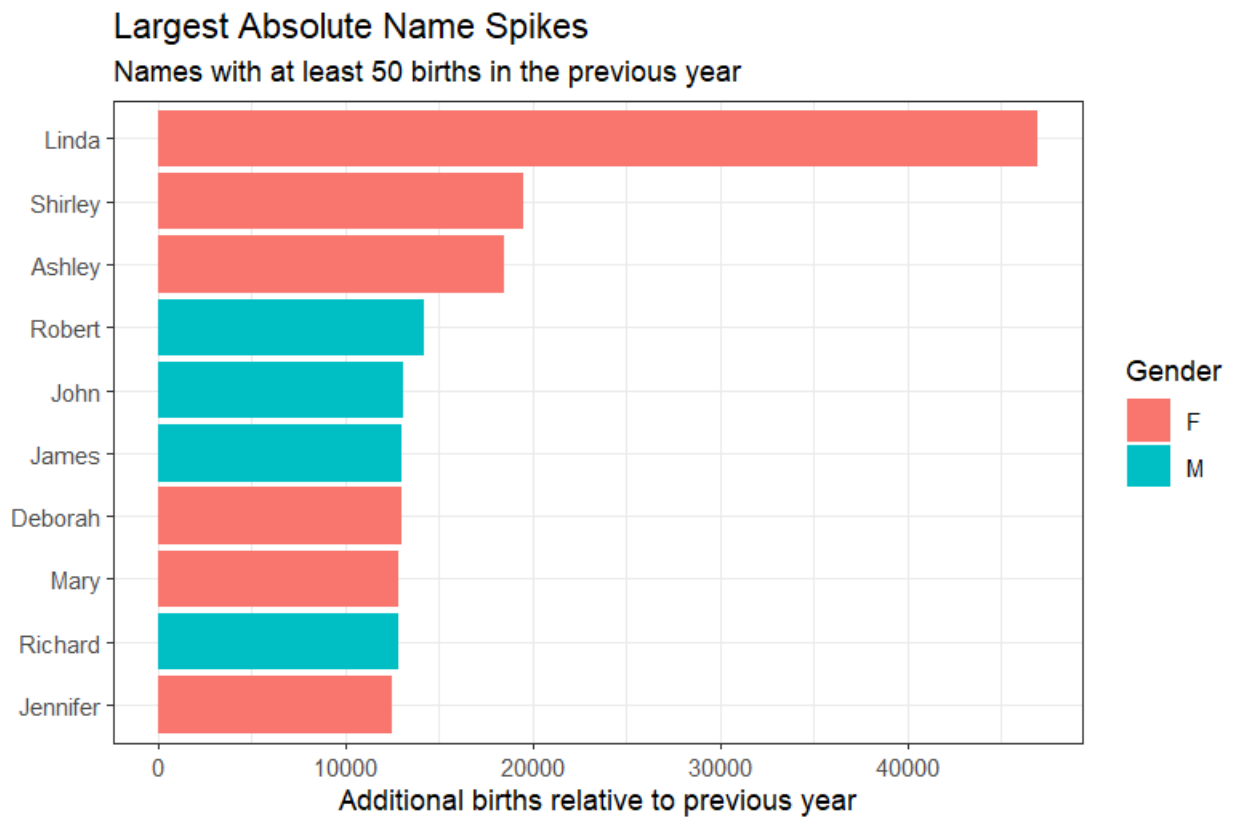


Figure 4.2: Largest absolute increases in baby-name popularity.

4.2.1. Key Findings

- Absolute spikes identify names that achieved widespread adoption.
- These names reflect broad national popularity rather than niche trends.
- Large absolute increases often indicate substantial cultural influence.

5. Media Influence on Naming

5.1. Case Study: Whitney

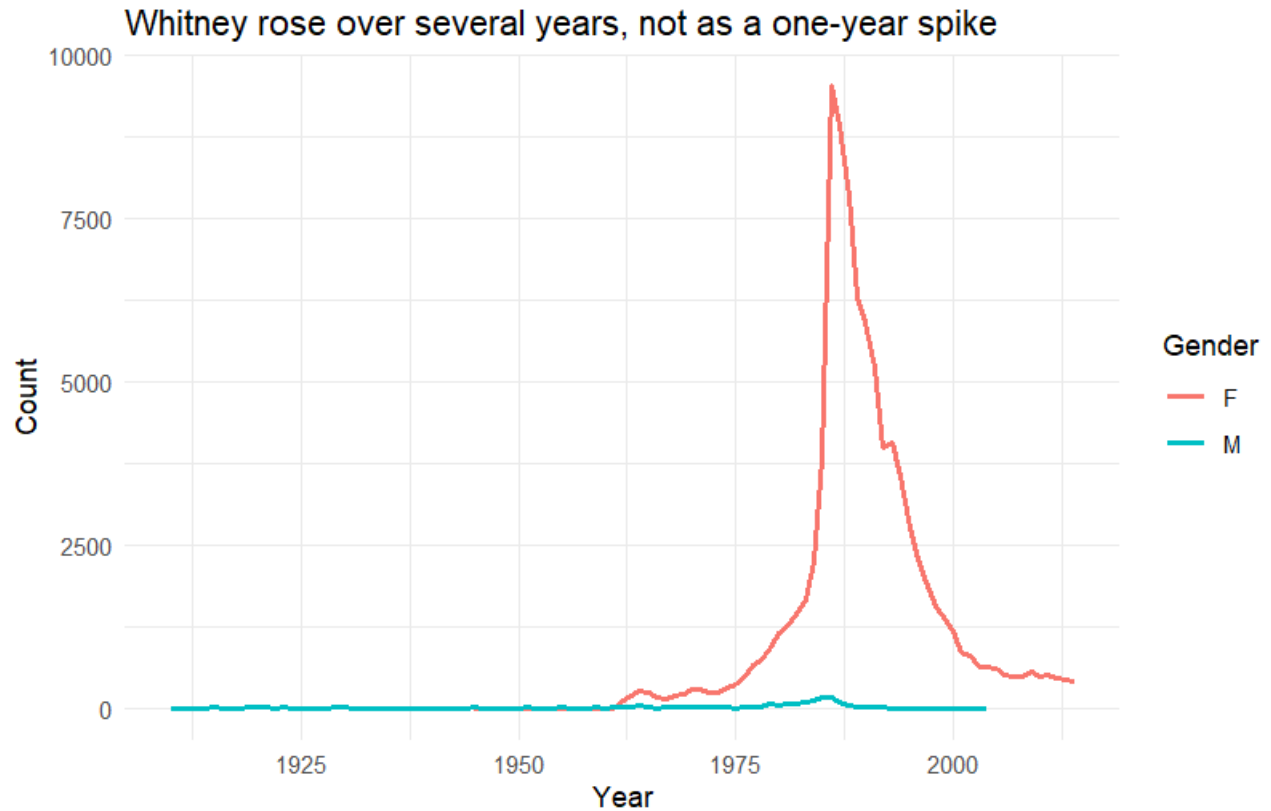


Figure 5.1: Popularity of the name Whitney through time.

5.1.1. Key Findings

- Whitney displays sustained growth during the 1980s.
- The increase coincides with the popularity of Whitney Houston.
- The pattern reflects long-term cultural influence rather than a one-year fad.

5.1.2. Business Implication

- Celebrity-associated names can generate lasting public recognition.
- Names with enduring cultural relevance may provide stronger branding opportunities.

6. Long-Term Naming Trends

6.1. Evolution of Selected Names

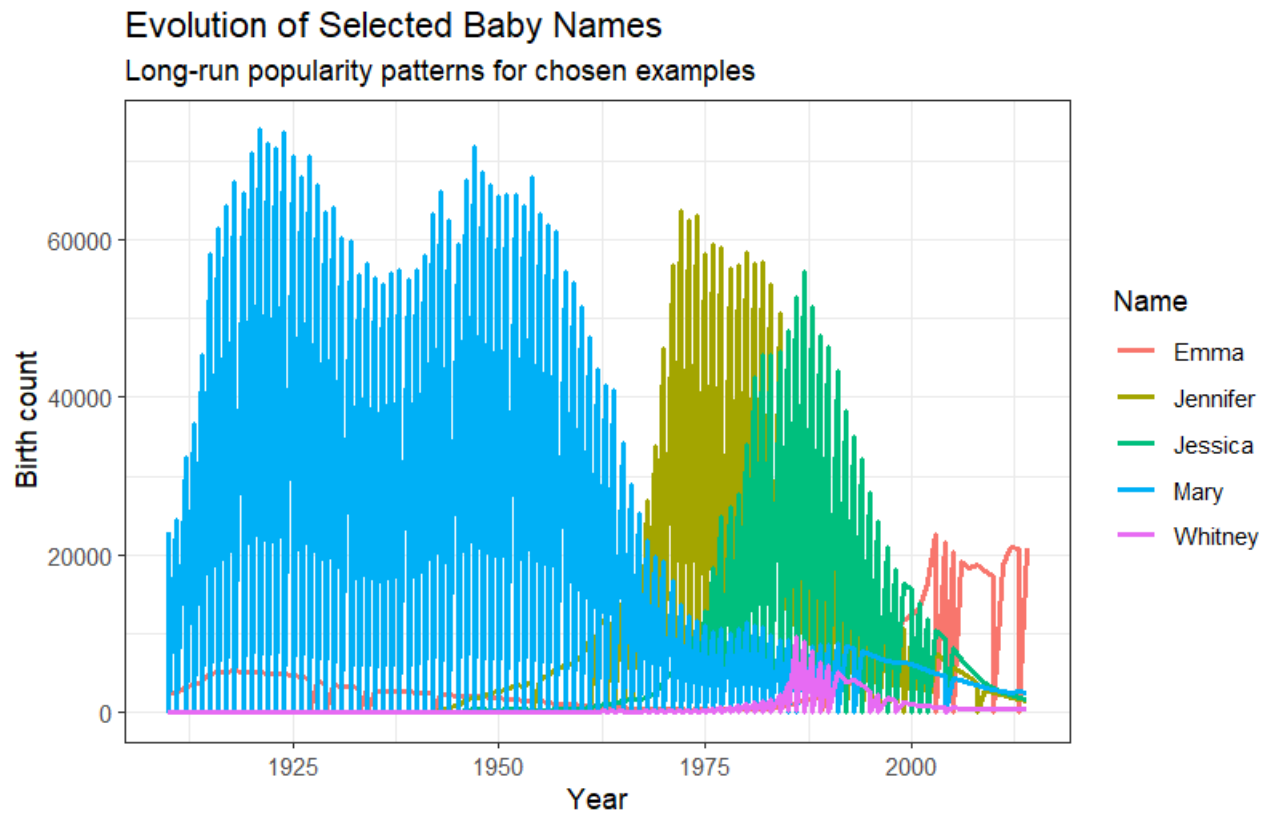


Figure 6.1: Evolution of selected baby names through time.

6.1.1. Key Findings

- Some names remain popular across multiple generations.
- Others exhibit distinct rise-and-fall cycles.
- Naming preferences evolve considerably through time.

6.2. Top Names by Decade

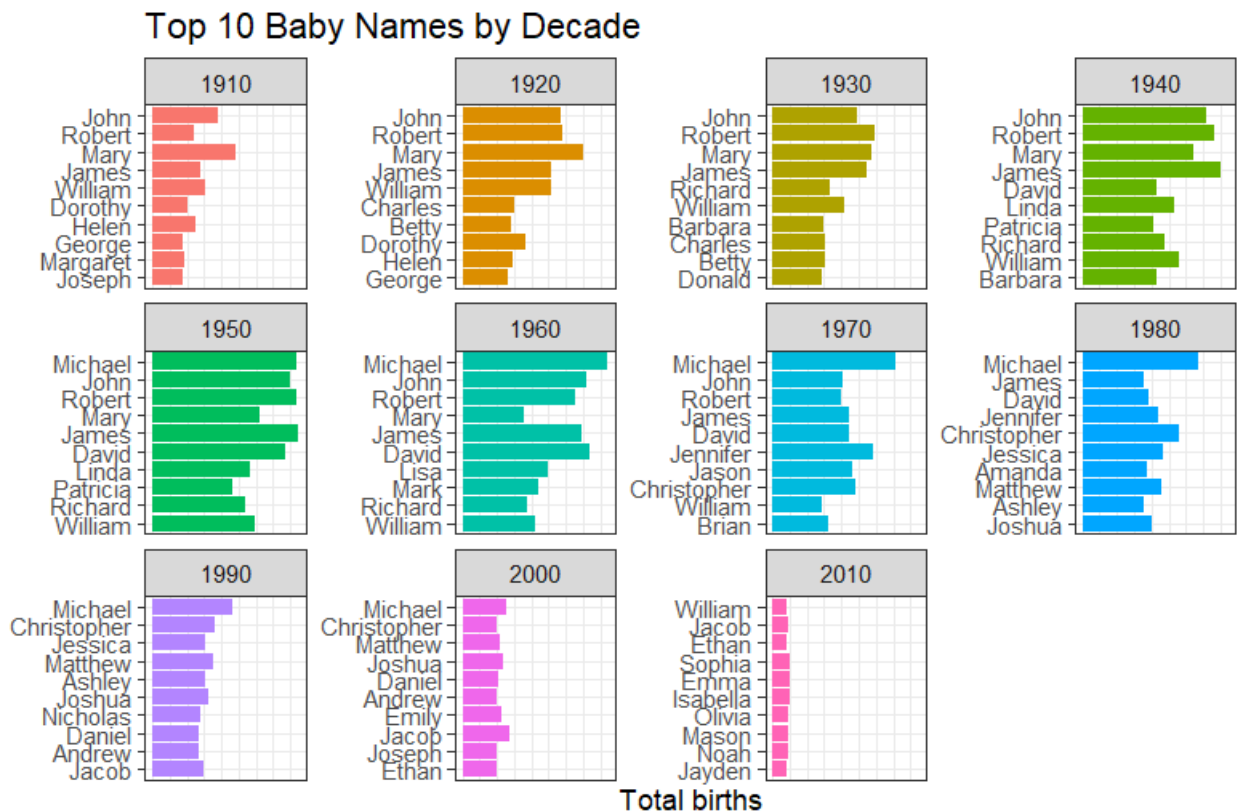


Figure 6.2: Top baby names by decade.

6.2.1. Key Findings

- Distinct naming fashions emerge in different decades.
- Some names dominate only a single generation.
- Other names demonstrate remarkable longevity.

6.3. Decade Heatmap

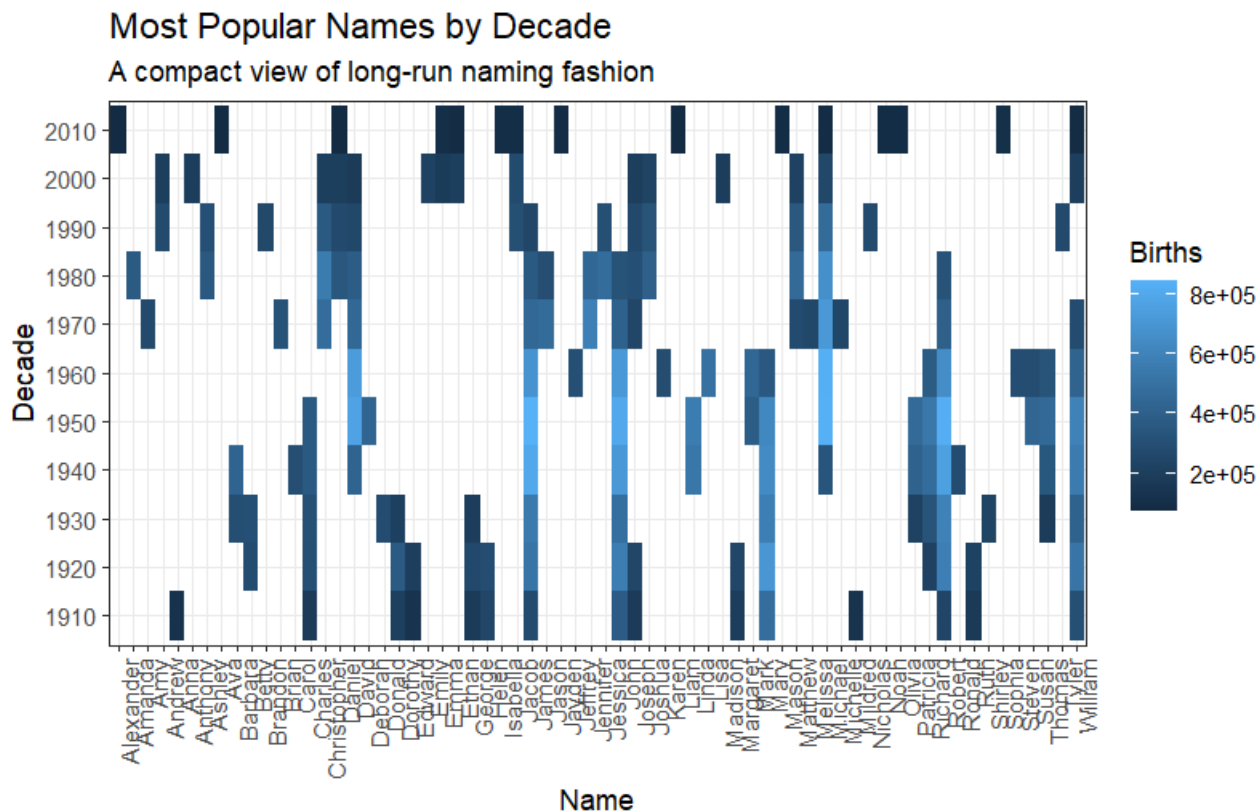


Figure 6.3: Heatmap of the most popular names by decade.

6.3.1. Key Findings

- The heatmap highlights changing naming fashions over time.
- Long-lived names remain visible across multiple decades.
- Most names experience finite popularity cycles.

7. New York Toy Character Name Analysis

7.1. Recommended Toy Names

A toy-name score was developed using popularity, longevity and stability within New York State.

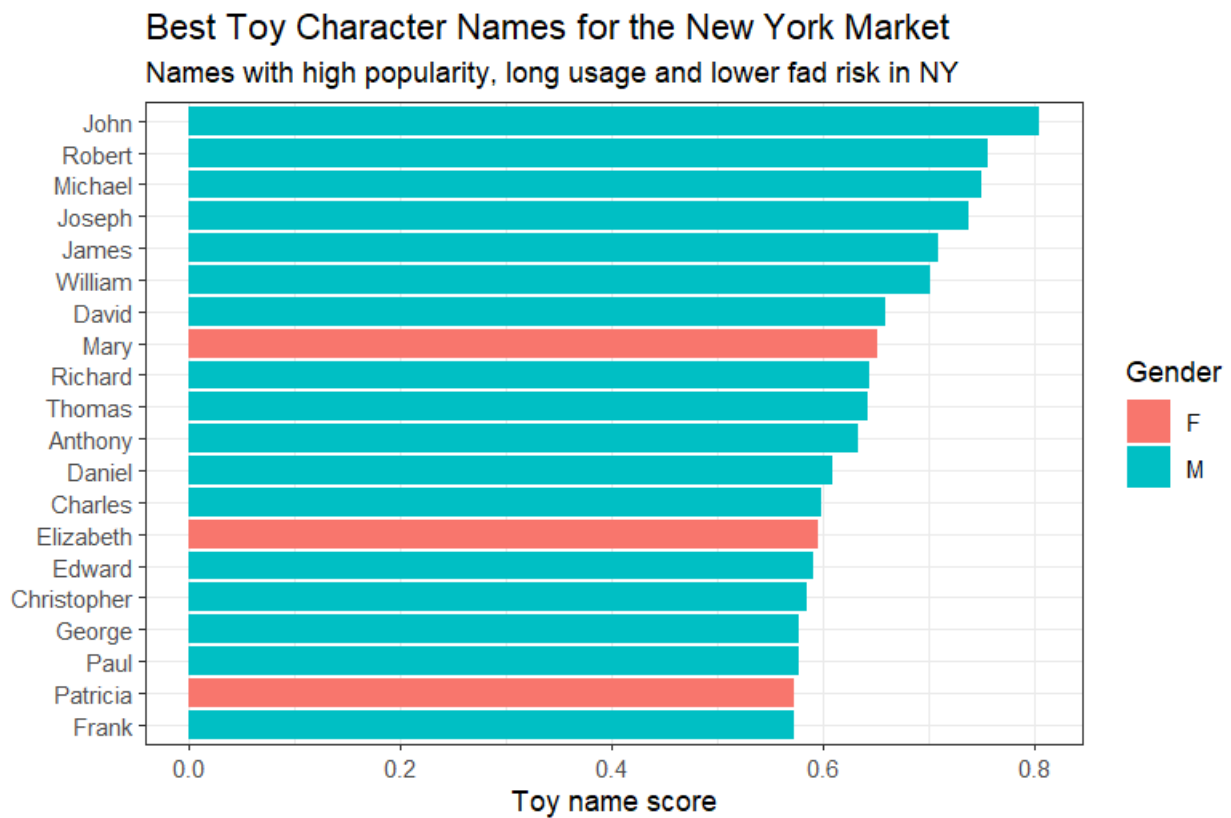


Figure 7.1: Highest scoring toy character names in New York.

7.1.1. Key Findings

- Recommended names combine popularity and long-term stability.
- These names are familiar to multiple generations of consumers.
- Such names represent lower branding risk.

7.2. High-Risk Toy Names

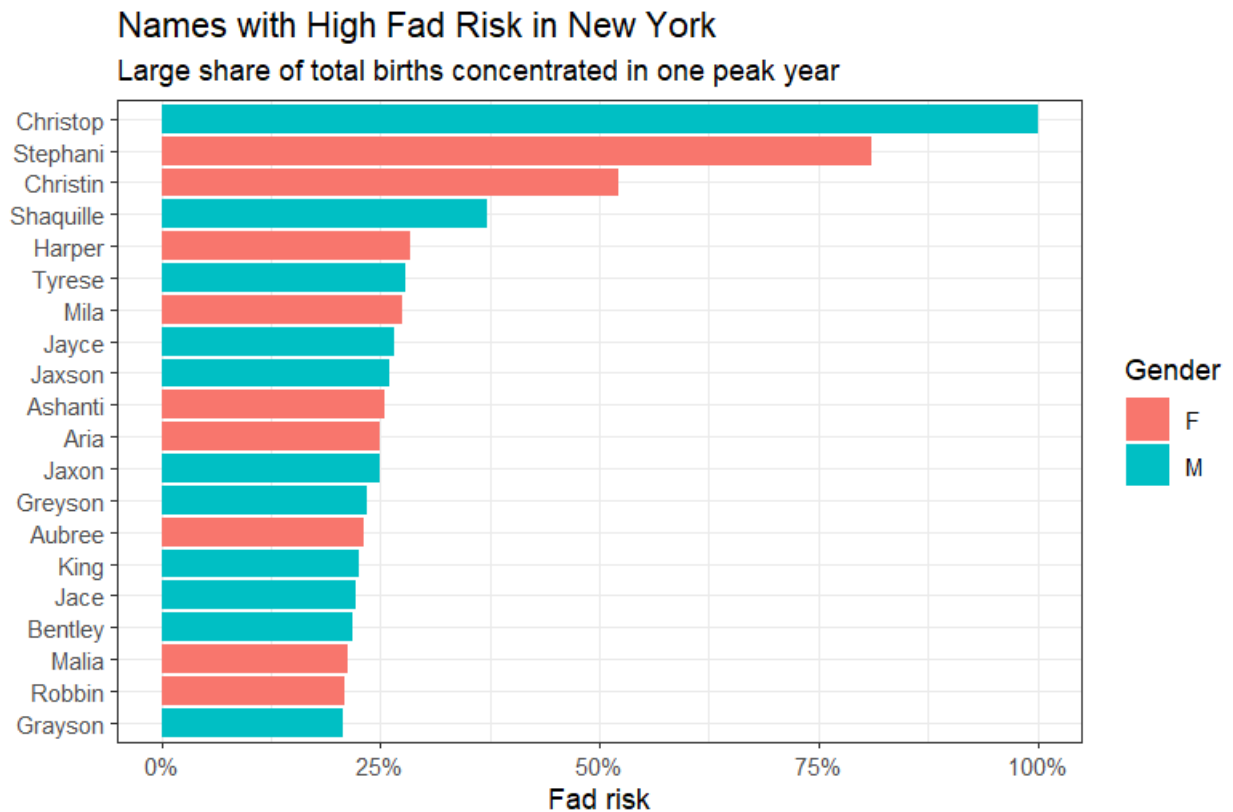


Figure 7.2: Names exhibiting high fad risk in New York.

7.2.1. Key Findings

- These names concentrate popularity within a short period.
- They are more vulnerable to changing trends.
- Such names may be suitable for short-term products but not flagship brands.

8. Recommendations

8.1. Long-Term Toy Franchises

Recommended characteristics:

- High persistence

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- High popularity
 - Low volatility
 - Multi-generational familiarity

Benefits:

- Lower branding risk
- Greater longevity
- Broader market appeal

8.2. Trend-Based Product Lines

Recommended characteristics:

- Strong recent growth
- Association with celebrities, music or television

Benefits:

- Immediate recognition
- Potential short-term sales boost

Risks:

- Higher probability of becoming outdated

9. Conclusion

Analysis of US baby naming trends indicates that naming persistence has declined since approximately 1990, resulting in shorter-lived naming fashions. Media and celebrity influences continue to shape naming behaviour and can generate substantial popularity increases. For a New York toy design agency, names demonstrating long-term stability and broad familiarity provide the strongest foundation for sustainable toy branding, while highly volatile names are better suited to short-term marketing opportunities.